8.1 Stages of Compilation

The C compilation process consists of preprocessing, compilation, assembling, and linking.

- Preprocessing: "gcc -E task8.c" expands macros and processes header files.
- Compilation: "gcc -S task8.c" generates assembly code.
- Assembling: "gcc -c task8.c" generates an object file.
- Linking: "gcc task8.c -o task8" creates the executable program.

8.2 Role of the Preprocessor

The preprocessor expands "#define" macros, includes header files, and processes conditional compilation directives such as "#ifdef DEBUG".

8.3 Types of Errors

- Compile-time error: Occurs when the compiler detects invalid syntax or incompatible code.
- Link-time error: Occurs when a required function or definition cannot be found during linking.
- Run-time error: Occurs while the program is executing, such as invalid input handling or division by zero.

8.4 Compiler Warnings

The command "gcc -Wall -Wextra" enables additional warnings. Warnings help identify possible errors, unused variables, and incorrect format specifiers. They should be investigated and corrected.

8.5 Alternative Concept and Complexity Analysis

A program split into multiple source files can be compiled separately. When one file changes, only that file and the linking step may need to be repeated. This can reduce compilation time for large projects, although multiple object files require additional storage.

```
#include <stdio.h>

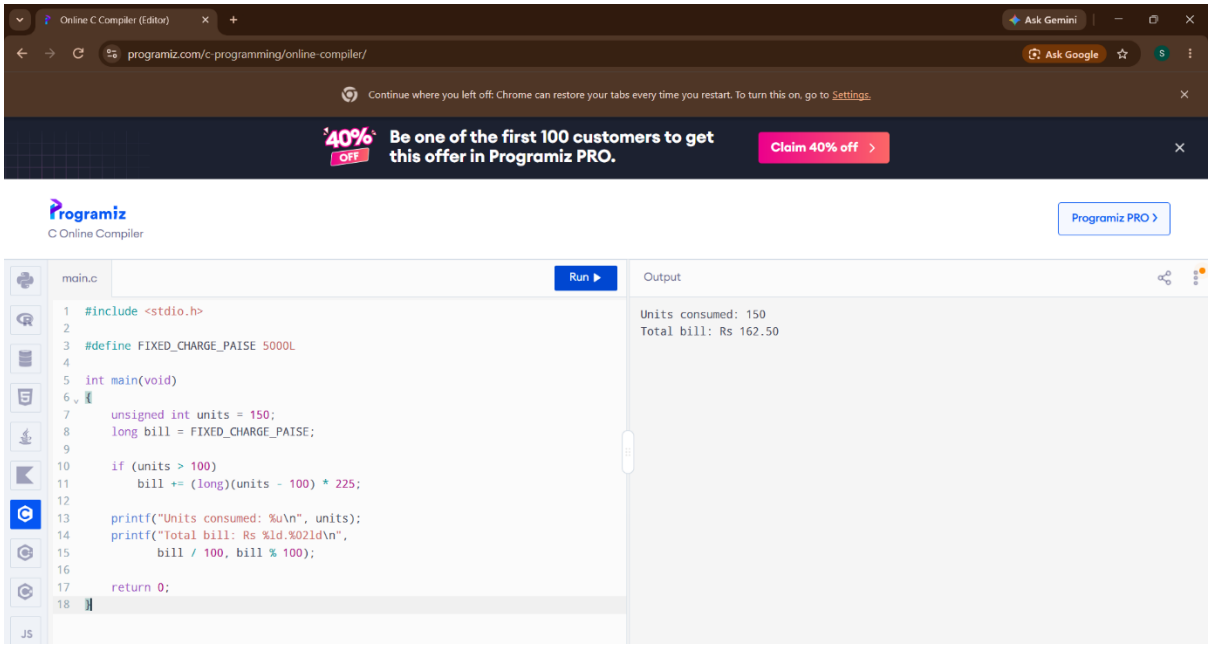
#define FIXED_CHARGE_PAISE 5000L

int main(void)
{
    unsigned int units = 150;
    long bill = FIXED_CHARGE_PAISE;

    if (units > 100)
        bill += (long)(units - 100) * 225;

    printf("Units consumed: %u\n", units);
    printf("Total bill: Rs %ld.%02ld\n",
           bill / 100, bill % 100);

    return 0;
}
```


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main.c
Run
Output
#include <stdio.h>
#define FIXED_CHARGE_PAISE 5000L
int main(void)
{
unsigned int units = 150;
long bill = FIXED_CHARGE_PAISE;
if (units > 100)
bill += (long)(units - 100) * 225;
printf("Units consumed: %u\n", units);
printf("Total bill: Rs %ld.%02ld\n",
bill / 100, bill % 100);
return 0;
}
Units consumed: 150
Total bill: Rs 162.50